Secondary Saturation and Cusp Geometry at Cancelling Double Grazing

Supplement to
Cyclicity Six and Saturation Geometry at Cancelling Double Grazing

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Mathematical supplement

Abstract

This supplement resolves the corner of the principal-scale saturation domain where three of the four limiting roots tend to zero. We consider a semistable limit cycle with two cancelling quadratic grazings whose active sides agree under transverse transport, with positive first passage strength B and negative closing curvature. The strength sum and a secondary scale tend to zero independently. Necessary bounds hold in every admissible family. In a fixed full-rank unfolding, the closures of the rescaled six-cycle domains converge in Hausdorff distance to the coefficient set of cubics with three nonnegative roots of sum one; every compact subset of its interior is realized uniformly for all sufficiently small independent scale pairs. For positive closing offset $a = R(0)$, six-cycle sequences tending to the reference satisfy the sharp bound $\limsup a/(-b)^3 \le 4/(27B^2)$, where b is the multiplier defect. For positive offset we realize six simple cycles, with one negative and five positive fixed points on the transverse section. The triple-root endpoint continues to an ordinary cusp. Near this cusp, the common return interval contains four or six simple cycles off the fold curves, five distinct cycles on the fold arcs away from the cusp, one double, and four distinct cycles at the cusp, one triple.

Keywords. Grazing bifurcation; Poincaré return maps; cyclicity; multiple limit cycles; cusp bifurcations.

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1 Introduction

A semistable limit cycle of a continuous piecewise smooth planar system may have two spatially separated quadratic grazings, with both active sides agreeing after transport to one transverse section. Each tangency adds a term of order $q^{3/2}$ to the return map. When the leading coefficients of the two terms cancel and the closing curvature is nonzero, the displacement has a double zero at the reference parameter, but nearby return maps can have six simple fixed points. The companion paper [1] determines the cyclicity and the saturation domain at the principal scale. Here we study smaller scales at a boundary of that domain.

The positive cluster at the principal scale is described by a quartic with four nonnegative roots of sum one. When its quadratic coefficient tends to zero, three roots approach zero and the fourth approaches one. Their next coefficient region is cubic. A polynomial rescaling suggests this cubic, but does not control the return map: the secondary scale can vanish at any rate relative to the strength sum, and critical points can approach the moving second threshold. The proof below uses divisions made before either scale vanishes and slope estimates near that threshold.

We work with positive reference first passage strength B and negative reference closing curvature $-C$, where $B, C > 0$. In terms of the offset $a = R(0)$, the multiplier defect b, the

strength sum m and the threshold separation δ, the secondary weights are

$$m = \mu, \qquad b = -\mu^2\sigma, \qquad \delta = \mu^3\sigma^2 d, \qquad a = \mu^4\sigma^3\alpha.$$

Both μ and σ are positive and independently small. The limiting cubic is $z^3 - z^2 + (3Bd/2)z + \alpha$. Its coefficient region $\mathcal{C}_B$ consists of cubics with three nonnegative roots of sum one. Proposition 3.7 proves that this is exactly the secondary cluster set and the Hausdorff limit of the six-cycle domains. Necessity holds in every admissible family; realization uses the standard unfolding of [1].

Six-cycle configurations with positive offset occur on a finer offset scale. With $h = -b$, Proposition 3.2 proves the sharp necessary bound $\limsup a/h^3 \le 4/(27B^2)$. Proposition 4.1 realizes every compact interior set of the associated parameters by six simple cycles, one with $q < 0$ and five with $q > 0$.

At the triple-root endpoint $(\alpha, d) = (-1/27, 2/(9B))$, Theorem 5.2 constructs an ordinary cusp for every sufficiently small independent pair (μ, σ). Lemma 5.1 locates every other root in the same interval I. The three simple roots outside the cluster convert the local cubic counts into the complete four/six-cycle diagram, including its folds and cusp.

The $q^{3/2}$ passage and the composition with regular transitions are classical [2, 3]; we use the root-profile form established in [1, Section 2]. The Malgrange preparation theorem [4, Chapter IV, Section 2, Theorem 2.1 and Note (1), p. 95] is applied after the multiple root has been located away from both thresholds, where the displacement is smooth in q.

Section 2 recalls the setting and the results of [1] used here. Section 3 passes from the quartic corner to the cubic domain through divisions, necessary bounds and root charts. Section 4 treats configurations with positive offset. Section 5 first locates all roots in I and then unfolds the secondary triple root.

2 Setting and results from the companion paper

We recall the setting of [1] and the estimates used below.

2.1 Hypotheses and return-map controls

We use the admissible families of [1, Definition 2.1], with their two matched quadratic contact boxes, transverse ports and inactive regular connections; both active sides agree after transport to one transverse section. The regular transitions are increasing in positive penetration coordinates, with common positive derivative margins. On a fixed interval $I = (-q_0, q_0)$ the composition is the first return map: it makes one turn, with no additional seam crossing or earlier section hit. Its common open domains include the grazing thresholds and the closure of a slightly larger interval. The return tube and the transport through nonidentity connections are given in [1, Lemma 2.2 and Proposition 2.6].

In the transported transverse coordinate, put

$$Q(q) = \delta + \Gamma_2(\Gamma_1(q) - \delta), \qquad P(q) = R(Q(q)), \qquad D(q) = P(q) - q. \tag{2.1}$$

The passages are increasing C^1 diffeomorphisms and

$$\Gamma_\nu(q) = q + A_\nu q_+^{3/2} + \tfrac{3}{4}A_\nu^2 q_+^2 + q_+^{5/2} L_\nu(\sqrt{q_+}, \lambda), \qquad q_+ = \max(q, 0), \tag{2.2}$$

with smooth signed-root profiles. More precisely, $\Gamma_\nu(r^2) = g_\nu(r)^2$ for $r > 0$, $g_\nu(0) = 0$, $g_\nu'(0) = 1$ and $\iota g_\nu \iota = g_\nu^{-1}$, where $\iota(r) = -r$. Set

$$T_\nu(r) = g_\nu(r)/r, \qquad \ell_\nu(r) = T_\nu(r)^2 + \tfrac{r}{2}(T_\nu^2)'(r).$$

These profiles extend smoothly through zero and $\Gamma'_\nu(r^2) = \ell_\nu(r) = 1 + \frac{3}{2}A_\nu r + O(r^2)$. The return-map controls and the reference signs are

$$\begin{aligned} a = R(0), \qquad b = R'(0) - 1, \qquad m = A_1 + A_2, \qquad \delta; \\ a = b = m = \delta = 0, \qquad A_1 = -A_2 = B > 0, \qquad \tfrac{1}{2}R''(0) = -C < 0. \end{aligned} \tag{2.3}$$

We write p_0 for the reference value of the remaining parameters. Here B, C are fixed reference constants, whereas $c(\lambda) = R''_\lambda(0)/2$ and the individual passage strengths retain all parameter dependence.

For necessary estimates, the original parameters are denoted by p. [1, Lemma 2.7] supplies independent auxiliary m, δ, with the same closing map, smooth bounded profiles and the strength identity $A_1(p, m, \delta) + A_2(p, m, \delta) = m$; restriction gives the original family. Write $A(p) = A_1(p, 0, 0)$, $k(p) = R'_{p,0,0}(0) = 1 + b > 0$ and $c_0(p) = R''_{p,0,0}(0)/2$. When the controls are independent we write $p = (a, b, \xi)$; the closing jets $R(0) = a$ and $R'(0) = 1 + b$ are fixed as m, δ vary, although higher closing terms may depend on these controls.

Realization requires the map $\lambda \mapsto (a, b, m, \delta)$ to be a submersion [1, (2.12)]. We then use the standard unfolding $\Lambda(a, b, m, \delta) = \lambda(a, b, m, \delta; \xi_0)$ of [1], defined on a neighbourhood $\mathcal{V} \subset \mathbb{R}^4$ of zero, and the same interval I. Write $N_\lambda(I) = \#\{q \in I : D_\lambda(q) = 0\}$. Families with this property are constructed in [1, Proposition 6.1 and Theorem 6.3]. We follow the conventions of [1, Section 2]: state derivatives are taken at fixed parameters, and all profile bounds hold on common compact reductions.

2.2 Divisions, curvature and saturation

We record the results of [1] used below. The closing expansion is

$$R(y) = a + (1 + b)y + c(\lambda)y^2 + y^3 S(y, \lambda). \tag{2.4}$$

After one common reduction, choose $r_0 > 0$ and $v_0 = 1/4$. For $0 < r \le r_0$, $|v| \le v_0$ and (m, p) in the common small neighbourhood, put $q = r^2$ and $\delta = r^2 v$. Both passages are active, and [1, Lemma 2.8] gives profiles smooth in signed (r, m, v, p) with

$$Q - q = mr^3 + \tfrac{3}{4}m^2 r^4 + r^5 L_0 + \delta r H(r, m, v, p), \qquad H(0, 0, 0, p) = \tfrac{3}{2}A(p). \tag{2.5}$$

For either sign of q and δ, including the thresholds, the same lemma gives

$$|Q - q - mq_+^{3/2}| \le K\{m^2 q_+^2 + q_+^{5/2} + |\delta|\sqrt{q_+ + |\delta|} + |\delta| q_+^{3/2}\}. \tag{2.6}$$

By [1, Lemma 3.1 and Proposition 3.2], the zeros of P'' in the outer region lie on the separation graph (3.2). Its derivative is $\Delta_r = r\mathcal{W}$, with $\mathcal{W}_r$ having a uniform nonzero margin of sign c_0/A; thus it has at most one positive critical point, which is nondegenerate. At fixed parameters the curvature factorization is

$$P''(r^2) = r^{-3}\mathcal{U}(r, m, \delta/r^2, p)[\delta - \Delta(r, m, p)], \qquad \operatorname{sgn}\mathcal{U} = -\operatorname{sgn} A.$$

For $A > 0, c_0 < 0, m > 0$, the critical point satisfies $r_c = -km/(4c_0) + O(m^2)$. [1, Lemma 3.3] covers the threshold layer, uniformly up to the thresholds. For positive separation the thresholds are 0 and $a_\delta = \Gamma_1^{-1}(\delta)$; in the threshold layer the signs of P'' where no passage, only the first passage, and both passages are active are $-, +, -$. For any fixed $L \ge 2$, these signs hold for $q \le L\delta$ away from the thresholds. In the active regions, $|Q''| \ge \kappa_L \delta^{-1/2}$, uniformly in the one-sided approach to each threshold. Together with the graph, they give the bound of six fixed points on I [1, Proposition 3.4].

If the bound is attained, [1, Lemma 4.2] gives $m, \delta, h = -b > 0$, two simple zeros $\zeta_1 < \zeta_2$ of P'' in the outer region, and the slopes recorded in (3.1). The five curvature blocks have

signs $-,+,-,+,-$ and each contains one zero t_j of D_q, with successive extremal values of signs $+,-,+,-,+$. Every fixed point is simple as a C^1 zero. Finally [1, Proposition 4.3] gives the weighted confinement

$$0 < h \le Km^2, \qquad 0 < \delta \le Km^3, \qquad |a| \le Km^4, \qquad \max_{D(q)=0} |q| \le Km^2. \tag{2.7}$$

The next section refines this confinement by evaluating the slope on the separation graph.

3 The two-scale secondary saturation domain

We determine the coefficient region reached when three roots of the principal quartic collapse to zero. A division of the slope gives the finer necessary scales; the joint return quotient and root charts then identify the region and realize its interior. Throughout this section we use the setting and conventions of Section 2.

3.1 The quartic corner and its cubic limit

The principal-scale coordinates of [1, Section 4.1] are

$$A_4 = \frac{C^3 a}{m^4}, \qquad S_4 = \frac{Ch}{m^2}, \qquad T_4 = \frac{3BC^2\delta}{2m^3}, \qquad x = \frac{C\sqrt{q}}{m}, \qquad h = -b.$$

Write e_j for the elementary symmetric polynomial of degree j, and set

$$\Phi_{A,S,T}(x) = A - Sx^2 + x^3 - x^4 + Tx, \qquad \mathcal{K}_4 = \{(-e_4, e_2, e_3) : r_i \ge 0,\ \textstyle\sum_{i=1}^4 r_i = 1\}.$$

Thus $\Phi_{A_4,S_4,T_4} = -\prod_{i=1}^4 (x - r_i)$ for a point in $\mathcal{K}_4$. We abbreviate this polynomial by Φ. The corner $S_4 \to 0$ of $\mathcal{K}_4$ explains the next scale. Order the four roots and write $u = r_1 + r_2 + r_3$, $v = r_1r_2 + r_1r_3 + r_2r_3$, $w = r_1r_2r_3$. Since $r_4 = 1 - u$,

$$S_4 = (1-u)u + v, \qquad T_4 = (1-u)v + w, \qquad -A_4 = (1-u)w.$$

Along any sequence with $S_4 \to 0^+$, the largest root tends to one and $u/S_4 \to 1$. After passing to a subsequence, $r_i/S_4 \to z_i \ge 0$ for $i = 1, 2, 3$, with $\sum_{i=1}^3 z_i = 1$. Along this subsequence, $(A_4/S_4^3, T_4/S_4^2) \to (-e_3(z), e_2(z))$. Conversely, taking small roots tz_i with $\sum z_i = 1$ and large root $1 - t$ realizes every such limit; here $S_4 = t - t^2(1 - e_2(z)) \sim t$.

Set $\sigma_4 = S_4$ and write $A_4 = \sigma_4^3\alpha$, $T_4 = \sigma_4^2 T_3$, and $x = \sigma_4 z$. Then

$$\Phi(\sigma_4 z)/\sigma_4^3 = \alpha - z^2 + z^3 + T_3 z - \sigma_4 z^4.$$

The polynomial identity identifies a candidate cubic. Transferring it to the return map when $h = -b = o(m^2)$ requires a division before either scale vanishes: an additive error $o(m^2)$ in the slope would not control h. The necessary estimates use sequences whose original parameters, including all remaining parameters, tend to the reference of Section 2.

3.2 Slope division and necessary bounds

The finer scales are forced by the slope at the first zero of P'' in the outer region and by the extrema of the displacement. We begin with a division of the slope on the separation graph.

For a return with six fixed points, let $a_\delta = \Gamma_1^{-1}(\delta)$ be the second threshold. By [1, Lemma 4.2] each open curvature block contains one zero t_j of D_q, with extremal signs $+,-,+,-,+$, and

$$D_q(0) < 0, \quad D_q(a_\delta) > 0, \quad D_q(\zeta_1) < 0, \quad D_q(\zeta_2) > 0. \tag{3.1}$$

Here $\zeta_1 < \zeta_2$ are the two zeros of P'' in the outer region; in particular $m, \delta, h = -b$ are positive.

The separation graph of [1, Section 3] is

$$\begin{aligned} \delta = \Delta(r, m, p) &= r^2\{mH_\Delta(r, m, p) + rB_\Delta(r, p)\}, \\ H_\Delta(0, 0, p) &= \frac{2}{A(p)}, \qquad B_\Delta(0, p) = \frac{16c_0(p)}{3k(p)A(p)}. \end{aligned} \tag{3.2}$$

Here B_Δ, H_Δ are smooth profiles.

Lemma 3.1 (Slope on the separation graph)**.** *After a common reduction there are jointly smooth profiles H_D, B_D such that*

$$D_q(r^2, p, m, \delta)\Big|_{\delta=\Delta(r,m,p)} = b + r\{mH_D(r, m, p) + rB_D(r, p)\}, \tag{3.3}$$

with

$$H_D(0, 0, p) = 3k(p), \qquad B_D(0, p) = 6c_0(p). \tag{3.4}$$

Proof. Use the signed-root profiles of [1, Section 3]. With $\delta = r^2 v$, the expression

$$D_q(r^2) = R'(Q)\ell_2(r\sqrt{T_1(r)^2 - v})\ell_1(r) - 1$$

extends smoothly in (r, m, v, p) on a common box. At $r = 0$ it equals b for all small m, v. Hadamard division in r therefore gives a smooth function $\mathcal{T}$ such that $D_q - b = r\mathcal{T}$. The first passage derivatives give

$$\ell_\nu(u) = 1 + \tfrac{3}{2}A_\nu u + O(u^2).$$

Consequently, at the root boundary,

$$\mathcal{T}(0, m, v, p) = \tfrac{3}{2}k(p)\{A_1(p, m, 0) + A_2(p, m, 0)\sqrt{1 - v}\}.$$

The aligned strength identity yields the linear terms $\frac{3}{2}km + \frac{3}{4}kAv$. On $m = v = 0$, finite cancellation gives $D_q - b = 2c_0r^2 + O(r^3)$. Joint Taylor division thus gives

$$\mathcal{T} = 2c_0 r + \tfrac{3}{2}km + \tfrac{3}{4}kAv + O((|r| + |m| + |v|)^2), \tag{3.5}$$

with a uniform smooth remainder.

Now substitute the graph $v = V(r, m, p) = mH_\Delta + rB_\Delta$ in $\mathcal{T}$, and call the resulting smooth function $\mathcal{T}_\Delta(r, m, p)$. Formula (3.5) gives

$$\mathcal{T}_\Delta(0, 0, p) = 0, \qquad \partial_m\mathcal{T}_\Delta(0, 0, p) = 3k, \qquad \partial_r\mathcal{T}_\Delta(0, 0, p) = 6c_0.$$

Define $B_D(r, p) = \mathcal{T}_\Delta(r, 0, p)/r$ by smooth division and

$$H_D(r, m, p) = \int_0^1 \partial_m\mathcal{T}_\Delta(r, sm, p)\, ds.$$

Thus $\mathcal{T}_\Delta = rB_D + mH_D$, proving (3.3)–(3.4). □

For $\delta > 0$, $q \geq 0$, the global passage estimate (2.6) gives

$$\begin{aligned} |D(q) - a| \leq K\{&hq + q^2 + mq^{3/2} + m^2q^2 + q^{5/2} \\ &+ \delta\sqrt{q + \delta} + \delta q^{3/2}\}, \qquad h = -b > 0. \end{aligned} \tag{3.6}$$

Indeed $0 \leq Q(q) \leq Kq$ uniformly. Below a_δ this follows from $Q = \Gamma_1(q)$; above a_δ, positivity and a common bound $\Gamma_2(y) \leq Ky$, with $K \geq 1$, give $\delta + \Gamma_2(\Gamma_1(q) - \delta) \leq K\Gamma_1(q)$. Insert this bound in the closing expansion (2.4) and apply (2.6).

Proposition 3.2 (Secondary necessary bounds)**.** *There is a constant K such that, for parameters sufficiently close to the reference in this sign case, every return with six fixed points satisfies, with $h = -b$,*

$$0 < h \le Km^2, \qquad 0 < \delta \le K\frac{h^2}{m}, \qquad -K\frac{h^3}{m^2} \le a \le Kh^3. \tag{3.7}$$

More precisely, with $k = 1 - h$ and the current first passage strength $A_1(\lambda)$,

$$a < \frac{4h^3}{27k^2A_1(\lambda)^2} + Kh^4. \tag{3.8}$$

Along every six-fixed-point sequence tending to the reference, this gives

$$\limsup \frac{a}{h^3} \le \frac{4}{27B^2}. \tag{3.9}$$

The leading term in (3.8) agrees with Simpson and Meiss's fold value $4(1-\Xi)^3/(27\gamma^2)$ for the truncated return [5, arXiv version, Eq. (72)], with $h = 1 - \Xi$ and $\gamma = kA_1(\lambda)$. Here the bound retains Kh^4 and is necessary for arbitrary six-cycle configurations; Proposition 4.1 proves sharpness by sequences approaching the endpoint.

Proof. [1, Proposition 4.3] gives $h \le Km^2$. Put $\rho_i = \sqrt{\zeta_i}$.

The root coordinate ρ_1 of the first curvature zero lies below the graph's unique positive critical point, whose expansion is

$$\rho_1 < r_c = -\frac{k(p)m}{4c_0(p)} + O(m^2).$$

Uniformly for $0 < r \le r_c$, the slope profiles satisfy

$$mH_D + rB_D \ge c_*m, \qquad c_* > 0.$$

To check the margin, divide by m and use (3.4): the limiting minimum on this interval is $3k + 6c_0[-k/(4c_0)] = 3k/2 > 0$. The negative slope at ζ_1 therefore yields

$$\rho_1 \le K\frac{h}{m}, \qquad \zeta_1 \le K\frac{h^2}{m^2}.$$

Using (3.2) at ρ_1, where $mH_\Delta + \rho_1 B_\Delta = O(m)$, proves $\delta \le Kh^2/m$.

For the upper offset bound, the negative minimum $t_2 = r_2^2$ lies where only the first passage is active, so that $Q = \Gamma_1(q)$. Uniform passage and closing expansions at fixed parameters give

$$D(r^2) = a - hr^2 + (1-h)A_1r^3 + O(r^4), \qquad D_q(r^2) = -h + \tfrac{3}{2}(1-h)A_1r + O(r^2).$$

The coefficient $\frac{3}{2}(1-h)A_1$ has a uniform positive lower bound. After reducing the state interval, the $O(r_2^2)$ term is absorbed in that linear term. Thus $D_q(t_2) = 0$ first gives $r_2 = O(h)$ and then

$$r_2 = \frac{2h}{3kA_1(\lambda)} + O(h^2).$$

The value expansion therefore yields

$$D(t_2) = a - \frac{4h^3}{27k^2A_1(\lambda)^2} + O(h^4). \tag{3.10}$$

Its strict negativity proves (3.8) and the coarse upper bound in (3.7). Since $k \to 1$ and $A_1(\lambda) \to B$ along a converging sequence, it also proves (3.9). Thus $a \le 4h^3/(27B^2) + o(h^3)$.

For the lower bound use the third extremum $t_3 \in (a_\delta, \zeta_1)$. It has $D(t_3) > 0$ and $t_3 \le Kh^2/m^2$. Substitute this and $\delta \le Kh^2/m$ in (3.6). The terms ht_3 and $mt_3^{3/2}$ are $O(h^3/m^2)$. The term t_3^2 is bounded by that size times h/m^2, which is bounded by the first estimate. The higher aligned terms are smaller. Finally,

$$\delta\sqrt{t_3+\delta} \le K\left(\frac{h^3}{m^2} + \frac{h^3}{m^{3/2}}\right).$$

The remaining separation term is smaller as well. Thus $|D(t_3) - a| \le Kh^3/m^2$, proving the lower offset bound.

□

3.3 The joint secondary quotient

In the standard unfolding introduce

$$m = \mu, \qquad b = -\mu^2\sigma, \qquad \delta = \mu^3\sigma^2 d, \qquad a = \mu^4\sigma^3\alpha, \qquad \mu, \sigma > 0. \tag{3.11}$$

These weights give $\sigma_4 = S_4 = C\sigma$ and $T_3 = 3Bd/2$. The weight map has determinant $-\mu^9\sigma^5$ with rows (m, b, a, δ) and columns (μ, σ, α, d). Here $h = \mu^2\sigma$ and the secondary positive state coordinate is

$$q = \mu^2\sigma^2 z^2.$$

The threshold layer has relative size $\delta/q = \mu d/z^2$, so on fixed positive z-intervals both passages are active for small μ, even when σ tends to zero independently.

Definition 3.3 (Secondary saturation domain)**.** Use the neighbourhood $\mathcal{V}$, the standard unfolding Λ and the count $N_\lambda(I)$ of Section 2. Let $w_{\mu,\sigma}(\alpha, d) = (a, b, m, \delta)$ be given by (3.11). The secondary saturation domain is

$$\mathscr{S}_{\mu,\sigma} = \{(\alpha, d) \in \mathbb{R}^2 : w_{\mu,\sigma}(\alpha, d) \in \mathcal{V}, N_{\Lambda(w_{\mu,\sigma}(\alpha,d))}(I) = 6\}.$$

Define the compact set

$$\mathcal{C}_B = \left\{\left(-r_1r_2r_3, \frac{2(r_1r_2 + r_1r_3 + r_2r_3)}{3B}\right) : r_i \ge 0, \quad r_1 + r_2 + r_3 = 1\right\}. \tag{3.12}$$

We use Euclidean distance for the Hausdorff distance d_H between compact coefficient sets. The quotient below connects the cubic region to the return map, with no relation between μ and σ. Further charts and estimates of critical points then account for the remaining roots and the boundary.

Lemma 3.4 (Joint secondary quotient)**.** *On every compact positive z-interval and bounded set of rescaled parameters, the function*

$$\mathcal{F}_{\mu,\sigma}(z; \alpha, d) = \frac{D(\mu^2\sigma^2z^2)}{\mu^4\sigma^3}$$

extends jointly smoothly to signed (μ, σ) near $(0, 0)$. There is a jointly smooth profile E such that

$$\mathcal{F}_{\mu,\sigma} = \alpha - z^2 + z^3 + \tfrac{3}{2}Bdz - C\sigma z^4 + \mu E(\mu, \sigma, z, \alpha, d). \tag{3.13}$$

Every prescribed finite collection of mixed derivatives is bounded on closed reductions. The extension equals the rescaled displacement for $\mu, \sigma > 0$.

Proof. The root-profile division (2.5), before any weights are imposed, gives

$$Q - q = mr^3 + \tfrac{3}{4}m^2r^4 + r^5L_0 + \delta rH(r, m, v, p), \qquad v = \delta/r^2, \qquad H(0, 0, 0, p_0) = \tfrac{3}{2}B. \tag{3.14}$$

Set $r = \mu\sigma z$ and $v = \mu d/z^2$. Define

$$\begin{aligned} V &= z^3 + dzH + \mu\sigma^2 z^5 L_0 + \tfrac{3}{4}\mu^2\sigma z^4, \\ Y &= z^2 + \mu^2\sigma V. \end{aligned} \tag{3.15}$$

These are jointly smooth for positive z in a compact interval, including signed auxiliary μ, σ, and satisfy $Q - q = \mu^4\sigma^3 V$, $Q = \mu^2\sigma^2 Y$ for positive scales. The closing expansion gives

$$\mathcal{F}_{\mu,\sigma} = \alpha + V - Y + c(\lambda)\sigma Y^2 + \mu^2\sigma^3 Y^3 S(\mu^2\sigma^2 Y, \lambda). \tag{3.16}$$

At $\mu = 0$, all four controls in (3.11) vanish, for every small σ. Because ξ is fixed, $c = -C$ and $H = 3B/2$ there. The restriction of (3.16) to $\mu = 0$ is therefore exactly $\alpha - z^2 + z^3 + (3B/2)dz - C\sigma z^4$. Taylor integral division in μ proves (3.13) and the mixed bounds. □

For a dependent family whose original parameters tend to the reference, if $\mu_n, \sigma_n \to 0^+$ and (α_n, d_n) stay bounded, the same identity gives, on every compact positive z interval,

$$\frac{D_n(\mu_n^2\sigma_n^2 z^2)}{\mu_n^4\sigma_n^3} - \{\alpha_n - z^2 + z^3 + \tfrac{3}{2}Bd_n z\} \longrightarrow 0 \tag{3.17}$$

in every prescribed finite state derivative. Indeed (3.14) and (3.16) still hold along the sequence, with its original remaining parameters retained. Their profiles converge to their reference values, and all their z derivatives are controlled by the same smooth root expressions.

3.4 Charts for the remaining roots

The following charts supply the remaining roots both in the secondary interior and in the positive-offset configurations of Section 4.

Lemma 3.5 (Charts for the remaining roots)**.** *In the standard unfolding, for bounded α, d with $d > 0$, the following limits hold in C^1 on the indicated compact state intervals as $\mu, \sigma \to 0^+$ independently:*

$$\frac{D(\mu^2\sigma^2 x)}{\mu^4\sigma^3} \longrightarrow \alpha - x, \qquad x \Subset (-\infty, 0), \tag{3.18}$$

$$\frac{D(-hX/C)}{h^2/C} \longrightarrow X - X^2, \qquad X \Subset (0, \infty), \tag{3.19}$$

$$\frac{D(\mu^2 X)}{\mu^4} \longrightarrow X^{3/2} - CX^2, \qquad X \Subset (0, \infty). \tag{3.20}$$

Convergence is uniform when the rescaled parameters range over a compact set.

Proof. For $q < 0$ both passages are exactly inactive because $\delta > 0$. Thus (2.4) gives $D(q) = a + bq + c(\lambda)q^2 + q^3 S(q, \lambda)$. Substituting $q = \mu^2\sigma^2 x$ and $q = -hX/C$, using (3.11), $h = \mu^2\sigma$ and $c(\lambda) \to -C$, proves (3.18) and (3.19).

For the positive outer chart set $r = \mu\sqrt{X}$ and $v = \mu\sigma^2 d/X$ in the root division (3.14). Here $v \to 0$ uniformly on the prescribed positive compact intervals, so both passages are active in this chart. The division gives $(Q - q)/\mu^4 \to X^{3/2}$ and $Q/\mu^2 \to X$ in C^1. In (2.4), $a/\mu^4 = \sigma^3\alpha$, $b/\mu^2 = -\sigma$, $c(\lambda) \to -C$, and the cubic closing remainder divided by μ^4 tends to zero. This proves (3.20). Smooth profiles give the stated C^1 uniformity on compact sets of states and rescaled parameters as $\mu, \sigma \to 0^+$ independently. □

3.5 Critical points and the limiting cubic

For the necessary limits we again allow dependent controls. For $d \geq 0$, write $T_3 = 3Bd/2$ and

$$F_{\alpha,d}(z) = z^3 - z^2 + T_3 z + \alpha.$$

Necessity requires control at the left endpoint of the positive root chart. We first place the third and fourth extrema on the secondary scale. For a positive limit of d, separated passage slopes exclude the threshold layer; when $d \to 0$, the third maximum supplies the missing value estimate.

Lemma 3.6 (Limiting critical points)**.** *Let a sequence in an admissible family tend to its reference, with six distinct fixed points, $\mu = m \to 0^+$ and $\sigma = -b/m^2 \to 0^+$. Pass to a subsequence on which the bounded normalized parameters satisfy $(\alpha, d) \to (\alpha_0, d_0)$. If $d_0 > 0$, the third and fourth displacement extrema satisfy*

$$\frac{\sqrt{t_3}}{\mu\sigma} \longrightarrow z_- = \frac{1 - \sqrt{1 - 3T_{3,0}}}{3}, \qquad \frac{\sqrt{t_4}}{\mu\sigma} \longrightarrow z_+ = \frac{1 + \sqrt{1 - 3T_{3,0}}}{3}, \qquad T_{3,0} = \tfrac{3}{2}Bd_0,$$

where $0 < T_{3,0} \leq 1/3$. Moreover,

$$F_{\alpha_0,d_0}(z_-) \geq 0, \qquad F_{\alpha_0,d_0}(z_+) \leq 0, \qquad \alpha_0 \leq 0. \tag{3.21}$$

If $d_0 = 0$, then $\alpha_0 = 0$.

Proof. Proposition 3.2 gives

$$-K \leq \alpha \leq K\mu^2, \qquad 0 < d \leq K.$$

Write $\rho_1 = \mu\sigma z_c$. The same bounds control z_c. The graph and slope identities yield

$$d = z_c^2\{H_\Delta + \sigma z_c B_\Delta\} = \frac{2}{B}z_c^2 + o(1), \qquad \frac{D_q(\zeta_1)}{h} = -1 + z_c H_D + \sigma z_c^2 B_D = -1 + 3z_c + o(1). \tag{3.22}$$

The strict negative slope gives $\limsup z_c \leq 1/3$ and therefore $d_0 \leq 2/(9B)$.

We already have $0 < t_3 < \zeta_1 = O(\mu^2\sigma^2)$. To obtain the same bound for t_4, first note that

$$\rho_2/\mu \longrightarrow \frac{3}{8C}. \tag{3.23}$$

In fact $\rho_2 = O(\mu)$ by the curvature graph, it is beyond the positive critical point $r_c/\mu \to 1/(4C)$ of the graph, and $\delta/\mu^3 = \sigma^2 d \to 0$. Dividing (3.2) by μ^3 shows that every limit w of ρ_2/μ satisfies

$$0 = w^2\left(\frac{2}{B} - \frac{16C}{3B}w\right).$$

The lower bound supplied by this critical point selects $w = 3/(8C)$. Choose a fixed Z larger than the bound for z_c and large enough that $-1 + 3Z/2 > 0$. By (3.17) and the chain rule in the state,

$$\frac{D_q(\mu^2\sigma^2Z^2)}{h} = -1 + \tfrac{3}{2}Z + \frac{3Bd}{4Z} + o(1) > 0.$$

For small scales this state lies in (ζ_1, ζ_2), by (3.23). The derivative is strictly increasing on that interval, so its unique zero t_4 lies before the test state. Thus both normalized critical states are bounded.

Suppose now that $d_0 > 0$. We exclude a zero of D_q at $q = \mu^2\sigma^2 z^2$ with $z \to 0$ where both passages are active. Put $y = \Gamma_1(q) - \delta \geq 0$. Along this sequence the parameters and state lie in the common small domain, $A_1 \geq B/2$, and $m, \delta > 0$. Since

$$\frac{q^{3/2}}{\delta} = \frac{\sigma z^3}{d} \longrightarrow 0,$$

the estimate of [1, (4.11)] applies and gives $D_q(q) \geq \kappa\delta/\sqrt{q} - K(h+q)$. With the exact substitutions

$$\frac{\delta}{\sqrt{q}\,h} = \frac{d}{z}, \qquad \frac{q}{h} = \sigma z^2,$$

we obtain, for a constant $c_* > 0$,

$$\frac{D_q(q)}{h} \geq -K + c_* \frac{d}{z} - K\sigma z^2. \tag{3.24}$$

The right side tends to $+\infty$, which contradicts a zero of D_q. The estimate also holds as $y \to 0$, at the second threshold.

The two normalized critical points now remain in a compact subset of $(0, \infty)$. Their limits are zeros of $F'_{\alpha_0,d_0} = 3z^2 - 2z + T_{3,0}$, by (3.17). Their order on either side of z_c, whose limit is $\sqrt{Bd_0/2} = \sqrt{T_{3,0}/3}$, selects z_-, z_+, respectively; at $T_{3,0} = 1/3$ the two limits agree. The signs $D(t_3) > 0$, $D(t_4) < 0$ give (3.21). The upper offset bound already gives $\alpha_0 \leq 0$.

If $d_0 = 0$, (3.22) instead gives $z_c \to 0$, so $z_M = \sqrt{t_3}/(\mu\sigma) \leq z_c \to 0$. At this state (3.6), divided by $\mu^4\sigma^3$, gives the explicit bound

$$\begin{aligned}\frac{|D(t_3) - a|}{\mu^4\sigma^3} \leq K\{&z_M^2 + z_M^3 + \sigma z_M^4 + \mu^2\sigma z_M^4 + \mu\sigma^2 z_M^5 \\ &+ dz_M + \sqrt{\mu}\, d^{3/2} + \mu^2\sigma^2 d z_M^3\}.\end{aligned}$$

Every term tends to zero. Positivity of $D(t_3)$ implies $\alpha_0 \geq 0$, and $\alpha \leq K\mu^2$ implies $\alpha_0 \leq 0$. Therefore $\alpha_0 = 0$. □

3.6 The complete corner domain

The limits of the critical points give necessity, while the joint quotient and the charts of Lemma 3.5 give interior realization. The interior of $\mathcal{C}_B$ consists precisely of parameters for which $F_{\alpha,d}$ has three distinct positive roots. Equivalently,

$$\mathcal{C}_B = \left\{ (\alpha, d) : \begin{array}{l} \alpha \leq 0, \quad d \geq 0, \\ T_3^2 - 4T_3^3 + 4\alpha - 18T_3\alpha - 27\alpha^2 \geq 0 \end{array} \right\}. \tag{3.25}$$

Proposition 3.7 (Secondary corner domain)**.** *In the fixed sign sector the following statements hold.*

1. *Let parameters in an admissible family tend to the reference. Suppose each return has six distinct fixed points and*

$$m \longrightarrow 0^+, \qquad \sigma = \frac{-b}{m^2} \longrightarrow 0^+.$$

Then the normalized parameters

$$\alpha = \frac{a}{m^4\sigma^3} = \frac{am^2}{h^3}, \qquad d = \frac{\delta}{m^3\sigma^2} = \frac{m\delta}{h^2}, \qquad h = -b, \tag{3.26}$$

are bounded, and every cluster point belongs to $\mathcal{C}_B$. In particular

$$\liminf \alpha \geq -\frac{1}{27}, \qquad \limsup \alpha \leq 0, \qquad 0 \leq \liminf d \leq \limsup d \leq \frac{2}{9B}. \tag{3.27}$$

If $a \geq 0$ eventually, the last upper bound improves to

$$\limsup d \leq \frac{1}{6B}. \tag{3.28}$$

2. *Assume [1, (2.12)] and use the standard unfolding. For every compact $\mathcal{K} \Subset \operatorname{int} \mathcal{C}_B$, there exist $\mu_0, \sigma_0 > 0$ such that, for every*

$$0 < \mu < \mu_0, \qquad 0 < \sigma < \sigma_0, \qquad (\alpha, d) \in \mathcal{K},$$

the parameters (3.11) give exactly six simple fixed points in I. If $r_1 < r_2 < r_3$ are the positive roots of $F_{\alpha,d}$, their states are, uniformly on $\mathcal{K}$,

$$\begin{aligned} q_{N,1} &= -\frac{h}{C}(1 + o(1)), & q_{N,2} &= \mu^2\sigma^2(\alpha + o(1)), \\ q_{P,j} &= \mu^2\sigma^2(r_j^2 + o(1)), \quad 1 \leq j \leq 3, & q_{P,4} &= \frac{\mu^2}{C^2}(1 + o(1)). \end{aligned} \tag{3.29}$$

Here $o(1) \to 0$ as $\mu + \sigma \to 0$, with the two positive scales independent.

3. *Every point of $\mathcal{C}_B$, including every boundary point, is the limit of normalized parameters along a sequence of returns with six fixed points in the standard unfolding.*

   The unrestricted bounds in (3.27) are sharp. At the top endpoint $d_0 = 2/(9B)$, necessarily $\alpha_0 = -1/27$, and the limiting cubic is $(z - 1/3)^3$.

   For the standard unfolding, $\mathscr{S}_{\mu,\sigma}$ has nonempty compact closure in one common rectangle for all sufficiently small independent positive μ, σ, and

$$d_H(\overline{\mathscr{S}_{\mu,\sigma}}, \mathcal{C}_B) \longrightarrow 0 \quad (\mu + \sigma \to 0). \tag{3.30}$$

Figure 1 depicts $\mathcal{C}_B$.

Proof of Proposition 3.7. Boundedness of α, d follows from Proposition 3.2, so consider any cluster point. For $d_0 > 0$, Lemma 3.6 gives a cubic with positive critical states $z_- \leq z_+$, nonnegative maximum, nonpositive minimum, and $F_{\alpha_0,d_0}(0) = \alpha_0 \leq 0$. Hence it has three nonnegative real roots, counted with multiplicity. This conclusion includes equality of critical values or $\alpha_0 = 0$ by continuity, or directly by the intermediate value theorem on the intervals $[0, z_-]$, $[z_-, z_+]$, and $[z_+, \infty)$. At $T_{3,0} = 1/3$ the two critical states coincide; their two value inequalities force a triple root there. For $d_0 = 0$ the limiting polynomial is $z^2(z - 1)$. In every case Vieta's identities give membership in $\mathcal{C}_B$.

The discriminant of $z^3 - z^2 + T_3 z + \alpha$ is the expression in (3.25). Conversely, that discriminant and $\alpha \leq 0$, $T_3 \geq 0$ give three real roots with sum one, nonnegative product, and nonnegative pair sum. If a root were negative, the product condition would allow us to write the two nonpositive roots as $-u, -v$, with $u > 0$, $v \geq 0$, and the third as $1 + u + v$. Their pair sum would be $-u^2 - uv - v^2 - u - v < 0$, a contradiction. This proves (3.25). Strict positivity and distinctness characterize the interior; a zero root or a repeated root is on its boundary.

For nonnegative r_i with sum one, $r_1 r_2 r_3 \leq 1/27$ and $r_1 r_2 + r_1 r_3 + r_2 r_3 \leq 1/3$. These inequalities prove (3.27). Equality in the second requires $r_1 = r_2 = r_3 = 1/3$ and gives the stated top endpoint. If $a \geq 0$ eventually, every cluster point has $\alpha_0 = 0$. Its polynomial is

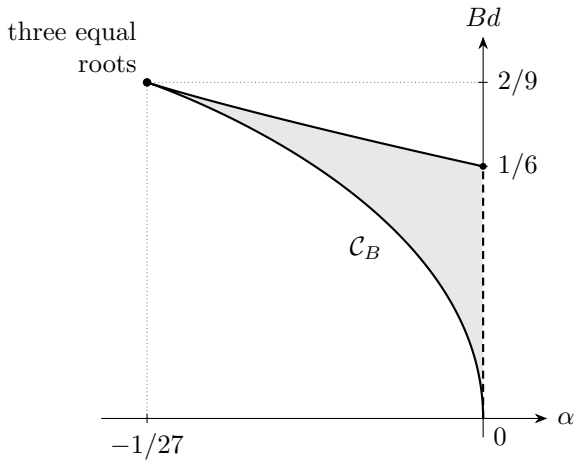


Figure 1: The secondary corner of the principal-scale saturation domain, in coordinates (α, Bd). The shaded interior corresponds to three distinct positive roots of $z^3 - z^2 + (3Bd/2)z + \alpha$. The solid boundary has a repeated root; the dashed segment has a zero root. The dot at $(-1/27, 2/9)$ has three equal roots. The solid curve is $(\alpha, Bd) = (-t^2(1-2t), 2(2t-3t^2)/3)$, $0 \le t \le 1/2$.

$z(z^2 - z + T_{3,0})$, whose two remaining roots are real and nonnegative only if $T_{3,0} \le 1/4$. This proves (3.28).

For the interior converse, fix a compact $\mathcal{K} \Subset \operatorname{int} \mathcal{C}_B$. The ordered positive roots r_j have uniform positive lower bounds, pairwise separation, and nonzero derivative margins. Lemma 3.4 transfers them to three simple roots on disjoint positive z-intervals for every independently small positive μ, σ, uniformly on $\mathcal{K}$. The other three roots follow from the charts of Lemma 3.5. The first chart has its negative root at $x = \alpha$, the second at $X = 1$, and the third at $X = 1/C^2$, all with strict uniform derivative margins. The negative secondary root and the three positive secondary roots have size $\mu^2\sigma^2$, whereas the outer negative root has size $\mu^2\sigma$ and the outer positive root size μ^2. The six isolating intervals are therefore disjoint for small σ. All their states belong to I for small scales.

The standard unfolding realizes (3.11) on one product $(0, \mu_0) \times (0, \sigma_0)$ for $\mathcal{K}$, and [1, Proposition 3.4] makes the six simple roots exhaustive; the same charts give (3.29).

For any boundary point, convexly perturb its ordered root triple toward $(1/6, 1/3, 1/2)$ to obtain interior parameters (α_j, d_j) tending to it. For the jth interior parameter choose positive $\mu_j, \sigma_j < 1/j$ below the two existence thresholds just proved. The corresponding return map has six simple roots, and its normalized parameters are exactly (α_j, d_j). This proves the diagonal assertion and, together with necessity, identifies the full cluster set as $\mathcal{C}_B$. It also proves sharpness of the unrestricted cluster bounds.

□

3.7 Hausdorff convergence

To obtain convergence of the full domains, we first place the normalized parameters of all six-cycle returns in one common rectangle. For the standard unfolding and the $\mathcal{V}, I$ of Definition 3.3, Proposition 3.2 bounds the full local domain by $-K \le \alpha \le K\mu^2$ and $0 < d \le K$. Choose $\mathcal{B} = [-M, M]^2$, $M > K$, containing $\mathcal{C}_B$ in its interior. The weighted parameters (3.11) tend uniformly to zero on this rectangle for independently small positive scales. Thus $\Lambda \circ w_{\mu,\sigma}$ is defined throughout $\mathcal{B}$ after one common reduction, and $\mathscr{S}_{\mu,\sigma} \subset \operatorname{int} \mathcal{B}$. The rectangle therefore contains the normalized parameters of every six-cycle return in $\mathcal{V}$, including those with positive

offset.

Completion of Proposition 3.7. We apply the compactness and finite-net argument that completes the proof of [1, Theorem 1.3] in [1, Section 4.1]. The common rectangle above makes every closure compact. An interior singleton of $\mathcal{C}_B$ is realized by part 2 of Proposition 3.7 for every sufficiently small independent scale pair, so these closures are nonempty.

For the upper inclusion, the weighted parameters tend uniformly to the fixed reference on $\mathcal{B}$, so part 1 applies along arbitrary sequences $\mu_n, \sigma_n \to 0^+$. For the lower inclusion, the root perturbation above gives interior density. Every finite interior net is compact, so part 2 realizes it in the standard unfolding on one product of positive scale intervals, with no relation between the scales. These are the two-scale inputs to the cited argument, which gives both Hausdorff inclusions. □

4 Optimal positive closing offset and its realization

Here positive offset means $a = R(0) > 0$. The constructions below use the finer offset scale $a = \eta h^3$, with $\alpha = \eta\mu^2 > 0$ and $\alpha \to 0$; the limiting cubic region $\mathcal{C}_B$ has $\alpha \le 0$. Two positive roots precede the second threshold and two persist on the secondary scale. The realization below adds the outer negative and positive roots to these two clusters; the bound of six then excludes further roots in I.

Proposition 4.1 (Positive-offset six-cycle returns)**.** *Assume [1, (2.12)] and use the standard unfolding on I. Put*

$$\eta_* = \frac{4}{27B^2}, \qquad d_* = \frac{1}{6B}.$$

For every compact $\mathcal{K} \Subset (0, \eta_*) \times (0, d_*)$, there are $\mu_0, \sigma_0 > 0$ such that the parameters

$$m = \mu, \qquad h = \mu^2\sigma, \qquad b = -h, \qquad \delta = \mu^3\sigma^2 d, \qquad a = \eta h^3 \tag{4.1}$$

give exactly one negative and five positive simple fixed points in I for every

$$0 < \mu < \mu_0, \qquad 0 < \sigma < \sigma_0, \qquad (\eta, d) \in \mathcal{K}.$$

Let $u_-(\eta) < u_+(\eta)$ be the two positive roots of $\eta - z^2 + Bz^3$, and put

$$r_\pm(d) = \frac{1 \pm \sqrt{1 - 6Bd}}{2}.$$

The six states satisfy, uniformly on $\mathcal{K}$,

$$\begin{aligned} q_N &= -\frac{h}{C}(1 + o(1)), \\ q_{F,\pm} &= h^2\big(u_\pm(\eta)^2 + o(1)\big), \\ q_{S,\pm} &= \mu^2\sigma^2\big(r_\pm(d)^2 + o(1)\big), \\ q_O &= \frac{\mu^2}{C^2}(1 + o(1)). \end{aligned} \tag{4.2}$$

Here $o(1) \to 0$ as $\mu + \sigma \to 0$, with the positive scales independent. The constants η_* in (3.9) and d_* in (3.28) are sharp: both are approached simultaneously by a sequence of these positive-offset returns.

Proof. Uniform convergence of (4.1) to zero on $\mathcal{K}$ lets the standard unfolding realize them on one product of scale intervals.

Below the second threshold $a_\delta = \Gamma_1^{-1}(\delta)$, the passage and closing expansions give

$$D(r^2) = a - hr^2 + kA_1(\lambda)r^3 + r^4T(r,\lambda), \qquad k = 1-h, \tag{4.3}$$

where T is jointly smooth on a common signed-root domain. For $r = hz$ on a compact positive z interval,

$$\frac{\Gamma_1(h^2z^2)}{\delta} = \frac{\mu z^2}{d}\{1 + O(h)\}.$$

Thus only the first passage is active at these states for all sufficiently small μ, uniformly on $\mathcal{K}$ and independently of σ. Their normalized displacement satisfies

$$\frac{D(h^2z^2)}{h^3} = \eta - z^2 + kA_1(\lambda)z^3 + hz^4T(hz,\lambda) = \eta - z^2 + Bz^3 + O_{C^1}(\mu). \tag{4.4}$$

Indeed the standard unfolding gives $A_1(\lambda) - B = O(\mu)$ on a bounded rectangle of scales, and the smooth root formula controls the displayed remainder and its z derivative.

The limiting polynomial is positive at zero and at infinity. Its positive critical point at $2/(3B)$ is a minimum, with value $\eta - \eta_* < 0$. It therefore has two positive simple roots $u_\pm(\eta)$.

The secondary weights (3.11) now have $\alpha = \eta\mu^2 > 0$. Substitution in the joint quotient (3.13) gives, on compact positive z intervals,

$$\frac{D(\mu^2\sigma^2z^2)}{\mu^4\sigma^3} = z(z^2 - z + \tfrac{3}{2}Bd) + O_{C^1}(\mu+\sigma). \tag{4.5}$$

For $0 < d < d_*$, its two nonzero limiting roots $r_\pm(d) > 0$ have derivatives $r_\pm(2r_\pm - 1) \neq 0$. Both passages are active on their isolating intervals because

$$\frac{a_\delta}{\mu^2\sigma^2z^2} = \frac{\mu d}{z^2}\{1 + o(1)\} \longrightarrow 0$$

uniformly on their isolating intervals.

Finally apply the outer charts (3.19) and (3.20) with $\alpha = \eta\mu^2$. Their C^1 limits $X - X^2$ and $X^{3/2} - CX^2$ have simple roots at 1 and $1/C^2$.

All four joint C^1 chart limits have compact isolating intervals with uniform derivative margins on $\mathcal{K}$. Simple-root continuation gives the six roots in (4.2) for common thresholds μ_0, σ_0, without a relation between the positive scales. The windows where only the first passage is active have size $h^2 = \mu^4\sigma^2$, the secondary windows size $\mu^2\sigma^2$, and the positive outer window size μ^2. Their successive ratios are $O(\mu^2)$ and $O(\sigma^2)$; the negative window is separated by sign. The second threshold lies strictly between the first-passage and secondary windows. All six windows lie in I for small scales, and [1, Proposition 3.4] excludes additional fixed points there. The charts also give (4.2).

For sharpness choose interior pairs (η_j, d_j) tending to (η_*, d_*). Apply the result to each singleton pair and choose positive $\mu_j, \sigma_j < 1/j$ below its two existence thresholds. The resulting returns have exactly

$$\frac{a_j}{h_j^3} = \eta_j \longrightarrow \eta_*, \qquad \frac{m_j\delta_j}{h_j^2} = d_j \longrightarrow d_*.$$

□

Corollary 4.2 (Six-cycle regions with positive offset)**.** *By [1, Corollary 3.6], in every admissible family the parameters giving six simple fixed points in I form an open set. Under [1, (2.12)], this set contains open subsets with $a > 0$, one negative fixed point and five positive fixed points, accumulating at the reference.*

Proof. The openness assertion is [1, Corollary 3.6]. For the returns of Proposition 4.1, shrink the neighbourhood to preserve $a > 0$ and the signs of the six simple roots. An interior parameter pair with both scales tending to zero gives accumulation at the reference. □

5 The secondary cusp on the whole interval

We first separate the secondary triple-root cluster from the three simple roots on the rest of the interval I. This localization supplies the complete cycle count when the cluster is unfolded.

5.1 Location of all roots near the secondary triple root

Lemma 5.1 (All roots near the secondary triple root)**.** *Use the standard unfolding. Fix a sufficiently small interval $J = (z_-, z_+) \ni 1/3$. There are a neighbourhood $\mathcal{U}$ of $(-1/27, 2/(9B))$, with compact closure in $\{\alpha < 0, d > 0\}$, and $\mu_0, \sigma_0 > 0$ such that the following statements hold for every*

$$(\alpha, d) \in \overline{\mathcal{U}}, \qquad 0 < \mu < \mu_0, \qquad 0 < \sigma < \sigma_0.$$

Then P'' has exactly two simple zeros $\zeta_1 < \zeta_2$ in the outer region, with

$$\sqrt{\zeta_1} = \mu\sigma\{\sqrt{Bd/2} + O(\mu + \sigma)\}, \qquad \sqrt{\zeta_2} = \mu\{3/(8C) + O(\mu + \sigma^2)\}. \tag{5.1}$$

The interval $\mathcal{J} = \{\mu^2\sigma^2 z^2 : z \in J\}$ contains ζ_1 and lies strictly between the second threshold and ζ_2. Outside $\mathcal{J}$ the displacement has exactly three roots, all simple, with uniform asymptotics

$$\begin{aligned} q_{N,1} &= -\frac{\mu^2\sigma}{C}(1 + o(1)), \quad q_{N,2} = \mu^2\sigma^2(\alpha + o(1)), \\ q_{P,\mathrm{out}} &= \frac{\mu^2}{C^2}(1 + o(1)). \end{aligned} \tag{5.2}$$

Inside $\mathcal{J}$ there are between one and three distinct roots. All limits are uniform as $\mu + \sigma \to 0$, without a condition on the ratio of the two scales.

Proof. Use the separation graph (3.2) with $p = (a, b, \xi_0)$. Substitution of $r = \mu\sigma z_g$ gives

$$d = z_g^2\{H_\Delta(\mu\sigma z_g, \mu, p) + \sigma z_g B_\Delta(\mu\sigma z_g, p)\}.$$

At zero scales the positive solution is $z_g = \sqrt{Bd/2}$, and the state derivative of the right side is $4z_g/B > 0$. Substitution of $r = \mu w_g$ instead gives

$$\sigma^2 d = w_g^2\{H_\Delta(\mu w_g, \mu, p) + w_g B_\Delta(\mu w_g, p)\}.$$

Its nonzero positive solution at zero scales is $w_g = 3/(8C)$; the state derivative there is $-3/(4BC)$. The parameter implicit function theorem proves (5.1) uniformly on a compact neighbourhood of the parameters. In the second equation the restriction at $\mu = 0$ depends on σ only through $\sigma^2 d$, giving the stated remainder. Both roots are in the outer region, since $\delta/\zeta_1 = O(\mu)$ and $\delta/\zeta_2 = O(\mu\sigma^2)$. The graph with its unique positive critical point and the threshold-layer estimate of [1, Lemma 3.3] exhaust the interval and exclude any other zero of P''. These two simple zeros and the threshold-layer signs give

q	$(-q_0, 0)$	$(0, a_\delta)$	(a_δ, ζ_1)	(ζ_1, ζ_2)	(ζ_2, q_0)
$\operatorname{sgn} D_{qq}$	$-$	$+$	$-$	$+$	$-$

where $a_\delta = \Gamma_1^{-1}(\delta)$.

The threshold identities $Q(a_\delta) = \delta$ and $a_\delta = \delta - A_1\delta^{3/2} + O(\delta^2)$ yield

$$D(a_\delta) - a = b\delta + A_1\delta^{3/2} + O(\delta^2), \qquad \frac{D(a_\delta)}{\mu^4\sigma^3} = \alpha + O(\sqrt{\mu}).$$

Indeed the three normalized errors are respectively $O(\mu)$, $O(\sqrt{\mu})$ and $O(\mu^2\sigma)$. Thus $D(0) < 0$ and $D(a_\delta) < 0$ uniformly. Strict convexity on $(0, a_\delta)$ excludes all roots between and at the thresholds.

Choose $J = (1/3 - \varepsilon, 1/3 + \varepsilon)$ with $\varepsilon > 0$ fixed and small. At the central parameters the zero-scale quotient is $(z - 1/3)^3$. Its endpoint values are $\pm\varepsilon^3$ and its endpoint derivatives are $3\varepsilon^2$. First shrinking $\mathcal{U}$, then the common scale rectangle, gives

$$\mathcal{F}(z_-) < 0 < \mathcal{F}(z_+), \qquad \mathcal{F}_z(z_\pm) > 0. \tag{5.3}$$

Write $q_\pm = \mu^2\sigma^2 z_\pm^2$. The curvature locations and $a_\delta \sim \delta$ give $a_\delta < q_- < \zeta_1 < q_+ < \zeta_2$. In particular,

$$D(q_-) < 0 < D(q_+), \qquad D_q(q_\pm) = \frac{\mu^2\sigma}{2z_\pm}\mathcal{F}_z(z_\pm) > 0.$$

On (a_δ, q_-) the derivative is decreasing and hence positive, so $D < D(q_-) < 0$. On (q_+, ζ_2) it is increasing and positive, so $D > D(q_+) > 0$, including $D(\zeta_2) > 0$.

Lemma 3.5 gives one simple root near each of $x = \alpha$, $X = 1$, and $X = 1/C^2$ in its three charts. Their nonzero limiting derivatives have common margins on $\overline{\mathcal{U}}$, and their isolators are disjoint for small σ. This proves (5.2). The negative half-interval is strictly concave and therefore has at most two zeros, so the two negative roots exhaust it. The outer positive root lies beyond ζ_2, since $1/C > 3/(8C)$, and has $D_q = \mu^2\{-1/(2C) + o(1)\} < 0$. Strict concavity on (ζ_2, q_0), starting at the positive value $D(\zeta_2)$, keeps D positive before this root and negative after it. This excludes every other root up to the fixed right endpoint. The negative-side concavity likewise covers its entire fixed tail.

Only $\mathcal{J}$ remains. Its endpoint values have opposite signs, and its only zero of P'' is ζ_1, with negative curvature before and positive curvature after it. Thus D_q has at most two zeros there, and Rolle's theorem gives at most three roots. All comparisons above use separate small bounds on μ and σ on the same interval. □

5.2 The cusp and its fold curves

Lemma 5.1 places the secondary cluster where both passages are active, with three simple roots on the rest of I. We apply the preparation theorem to the joint quotient of Lemma 3.4.

Theorem 5.2 (Cusp and cycle counts)**.** *For the standard unfolding, there are a neighbourhood $\mathcal{U} \ni (-1/27, 2/(9B))$ and $\mu_0, \sigma_0 > 0$ such that the following hold on $\mathcal{U}$ for every independent pair $0 < \mu < \mu_0$, $0 < \sigma < \sigma_0$. There is a unique nearby jointly smooth branch $(\alpha_c, d_c, z_c)(\mu, \sigma)$ of triple zeros of $\mathcal{F}_{\mu,\sigma}$ through $(-1/27, 2/(9B), 1/3)$ at zero scales. The separation $\delta_{\mathrm{cusp}} = \mu^3\sigma^2 d_c$ and state $q_{\mathrm{cusp}} = \mu^2\sigma^2 z_c^2$ give a limit cycle of multiplicity three.*

In the normalized variables $(\mu, \sigma, z, \alpha, d)$ there are jointly smooth changes of state and parameters, with a nonzero smooth factor, which reduce the local zero equation to

$$y^3 + uy + v = 0. \tag{5.4}$$

For each positive μ, σ, (u, v) are local coordinates on the slice $m = \mu$, $b = -\mu^2\sigma$ of the standard unfolding. Throughout $\mathcal{U}$, the prepared coordinates give the following counts in I: four simple cycles when $4u^3 + 27v^2 > 0$ and six simple cycles when $4u^3 + 27v^2 < 0$. Both regions are nonempty and open. On the fold arcs $4u^3 + 27v^2 = 0$, $(u, v) \neq (0, 0)$, there are five distinct cycles, one double; at $(u, v) = (0, 0)$ there are four distinct cycles, one triple.

In the rescaled parameters (α, d) the double-root curve is obtained by solving

$$\mathcal{F}(t;\alpha_f,d_f)=0, \qquad \mathcal{F}_z(t;\alpha_f,d_f)=0 \tag{5.5}$$

for (α_f, d_f), with t near $1/3$. The curve has an ordinary cusp at $t=z_c$ and consists of two ordinary fold arcs away from that point.

Proof. At $(\mu,\sigma,\alpha,d,z)=(0,0,-1/27,2/(9B),1/3)$, the quotient is $(z-1/3)^3$, and

$$D_{(\alpha,d,z)}(\mathcal{F},\mathcal{F}_z,\mathcal{F}_{zz})=\begin{pmatrix}1 & B/2 & 0\\ 0 & 3B/2 & 0\\ 0 & 0 & 6\end{pmatrix}, \qquad \det = 9B.$$

The parameter implicit function theorem gives the jointly smooth branch, which differs from its limiting value by $O(\mu+\sigma)$ and has $\mathcal{F}_{zzz}=6+O(\mu+\sigma)>0$. Moreover

$$\frac{q_{\text{cusp}}}{\delta_{\text{cusp}}}=\frac{z_c^2}{\mu d_c}=\frac{B}{2\mu}\{1+O(\mu+\sigma)\}\longrightarrow\infty.$$

Since the second threshold is asymptotic to δ_{cusp}, both passages are active at the triple root and D is smooth in q there; the state change $q=\mu^2\sigma^2z^2$ preserves multiplicity.

The Malgrange preparation theorem [4, Chapter IV, Section 2, Theorem 2.1 and Note (1), p. 95], with both scales as parameters, followed by removal of the quadratic term gives (5.4). At zero scales, with $\widehat{\alpha}=\alpha+1/27$ and $\widehat{d}=d-2/(9B)$,

$$\begin{aligned} u &= \tfrac{3}{2}B\widehat{d}+O((|\widehat{\alpha}|+|\widehat{d}|)^2),\\ v &= \widehat{\alpha}+\tfrac{B}{2}\widehat{d}+O((|\widehat{\alpha}|+|\widehat{d}|)^2).\end{aligned}$$

At zero scales the parameter Jacobian is $-3B/2\neq 0$, so it remains nonzero on a common reduction. Choose J of Lemma 5.1 inside the preparation chart, then reduce the parameters and the rectangle of scales so that every prepared cubic root pulls back into J. The cubic discriminant gives one or three simple roots on its two open sides, two distinct roots on a nonzero fold, and a triple root at the origin. Both open sides occur because the parameter change is a local diffeomorphism. Adding the three simple roots of Lemma 5.1 gives all stated counts on I.

For the fold curve, the equations $(\mathcal{F},\mathcal{F}_z)=0$ at $z=t$ have parameter Jacobian

$$D_{(\alpha,d)}(\mathcal{F},\mathcal{F}_z)=\begin{pmatrix}1 & 3Bt/2\\ 0 & 3B/2\end{pmatrix} \quad \text{at zero scales.}$$

The uniform implicit function theorem gives jointly smooth functions (α_f, d_f) on a compact t interval, with zero-scale curve $\alpha_0(t)=-t^2+2t^3$, $d_0(t)=2t(2-3t)/(3B)$. Differentiating the fold equations gives $\gamma_f'(z_c)=0$ for $\gamma_f=(\alpha_f,d_f)$, since their parameter Jacobian is invertible and $\mathcal{F}_z=\mathcal{F}_{zz}=0$ there. Smooth dependence on the scales and the zero-scale curve give

$$\det(\gamma_f''(z_c),\gamma_f'''(z_c))=\frac{48}{B}+O(\mu+\sigma)\neq 0. \tag{5.6}$$

Thus γ_f has an ordinary cusp. Along it, $\frac{d}{dt}\mathcal{F}_{zz}=\mathcal{F}_{zzz}>0$ at the cusp; after a common reduction all other points are ordinary double roots. □

Figure 2 shows the cycle counts on I in the prepared coordinates.

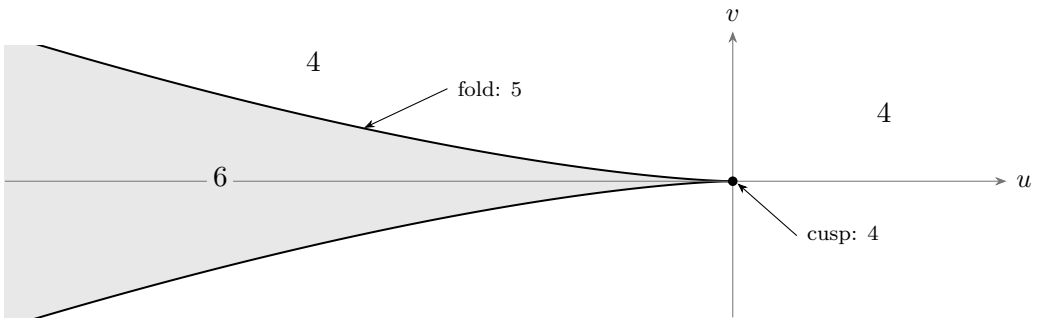


Figure 2: The cusp in the prepared coordinates of $y^3 + uy + v = 0$. The boundary is $4u^3 + 27v^2 = 0$: the shaded region has six simple cycles and the other region has four, including the three simple cycles outside the local cluster. An ordinary fold has five different cycles, one double; the cusp has four, one triple. A sufficiently small weighted rescaling of (u, v) places the diagram inside the common local parameter neighbourhood. The pullback of the discriminant to (α, d) is the local bifurcation curve, including its cusp.

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Data and code availability. This document is Online Resource 1. Its editable source and finite secondary polynomial checker accompany this paper. Online Resource 2 contains the separate symbolic verification package; Online Resource 3 contains computer-assisted proofs that, at each of two fixed parameter tuples, the harvesting model has at least six distinct simple periodic orbits. The analytic results are proved in the text; the computational files provide independent checks and fixed-parameter examples.

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